

Worksheet No. - 6

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Aim/Overview of the practical:

Learn how to create, query, and manage views in SQL to simplify database queries and provide a layer of abstraction for end-users.

Objective:

- **Data Abstraction:** To understand how to hide complex table joins and calculations behind a simple virtual table interface.
- **Enhanced Security:** To learn how to restrict user access to sensitive columns by providing views instead of direct table access.
- **Query Simplification:** To master the creation of views that pre-join multiple tables, making reporting easier for non-technical users.
- **View Management:** To understand the syntax for creating, altering, and dropping views, as well as the naming conventions required for efficient data access.
-

Input/Apparatus Used:

- PostgreSQL
- pgAdmin

Procedure/Algorithm/Code :

```
CREATE TABLE department (  
    dept_id INT PRIMARY KEY,  
    dept_name VARCHAR(50)  
);
```

```
CREATE TABLE employee (  
    emp_id INT PRIMARY KEY,
```

```
emp_name VARCHAR(50),  
  
salary NUMERIC,  
  
status VARCHAR(10),  
  
dept_id INT,  
  
FOREIGN KEY (dept_id) REFERENCES department(dept_id)  
  
);
```

```
INSERT INTO department VALUES
```

```
(1, 'IT'),  
(2, 'Sales'),  
(3, 'HR');
```

```
INSERT INTO employee VALUES
```

```
(1, 'Sanchit', 30000, 'Active', 2),  
(2, 'Swayam', 40000, 'Active', 2),  
(3, 'Anindita', 25000, 'Inactive', 1),  
(4, 'Ankush', 35000, 'Active', 3),  
(5, 'Roshan', 28000, 'Active', 1);
```

```
--1
```

```
CREATE VIEW active_employees AS  
  
SELECT emp_id, emp_name, dept_id  
  
FROM employee  
  
WHERE status = 'Active';
```

```
SELECT * FROM active_employees;
```

--2

```
CREATE VIEW employee_department_view AS
SELECT e.emp_id, e.emp_name, d.dept_name
FROM employee e
JOIN department d ON e.dept_id = d.dept_id;

SELECT * FROM employee_department_view;
```

--3

```
CREATE VIEW department_summary AS
SELECT d.dept_name,
       COUNT(e.emp_id) AS total_employees,
       AVG(e.salary) AS average_salary
FROM department d
JOIN employee e ON d.dept_id = e.dept_id
GROUP BY d.dept_name;

SELECT * FROM department_summary;
```

Output:

	emp_id [PK] integer	emp_name character varying (50)	salary numeric	status character varying (10)	dept_id integer
1	1	Sanchit	30000	Active	2
2	2	Swayam	40000	Active	2
3	3	Anindita	25000	Inactive	1
4	4	Ankush	35000	Active	3
5	5	Roshan	28000	Active	1

	dept_id [PK] integer	dept_name character varying (50)
1	1	IT
2	2	Sales
3	3	HR

	dept_name character varying (50)	total_employees bigint	average_salary numeric
1	Sales	2	35000.000000000000
2	IT	2	26500.000000000000
3	HR	1	35000.000000000000

	emp_id integer	emp_name character varying (50)	dept_name character varying (50)
1	1	Sanchit	Sales
2	2	Swayam	Sales
3	3	Anindita	IT
4	4	Ankush	HR
5	5	Roshan	IT

	emp_id integer	emp_name character varying (50)	dept_id integer
1	1	Sanchit	2
2	2	Swayam	2
3	4	Ankush	3
4	5	Roshan	1

Learning outcomes (What I have learnt):

1. I learned how to create and use views to simplify complex queries, making data retrieval easier and more structured for end-users.
2. I understood how views provide data abstraction by hiding underlying table structures and joins behind a virtual table interface.
3. I gained practical knowledge of implementing security through views, restricting access to sensitive columns and exposing only required data.
4. I developed the ability to design both simple and complex views, including filtered views, joined views, and aggregate summary views for real-world business scenarios.